Cybersecurity Lab Setup Report

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Subject: Cybersecurity Lab Setup

1. Objective

The objective of this lab was to set up a secure cybersecurity practice environment using virtual machines. The lab includes Kali Linux as the attacking machine, Metasploitable as the vulnerable target machine, and Wireshark for monitoring and analyzing network traffic.

2. Lab Environment

Component	Details
Host OS	Windows 11
Virtualization Software	VMware Workstation Pro
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2
Packet Analyzer	Wireshark
Network Mode	NAT / Host-Only

3. Virtual Machines

3.1 Kali Linux

Purpose

Kali Linux is a penetration testing distribution that contains numerous cybersecurity tools for ethical hacking, vulnerability assessment, and digital forensics.

Installed Tools

- Nmap
- Wireshark
- Metasploit Framework
- Burp Suite
- Hydra

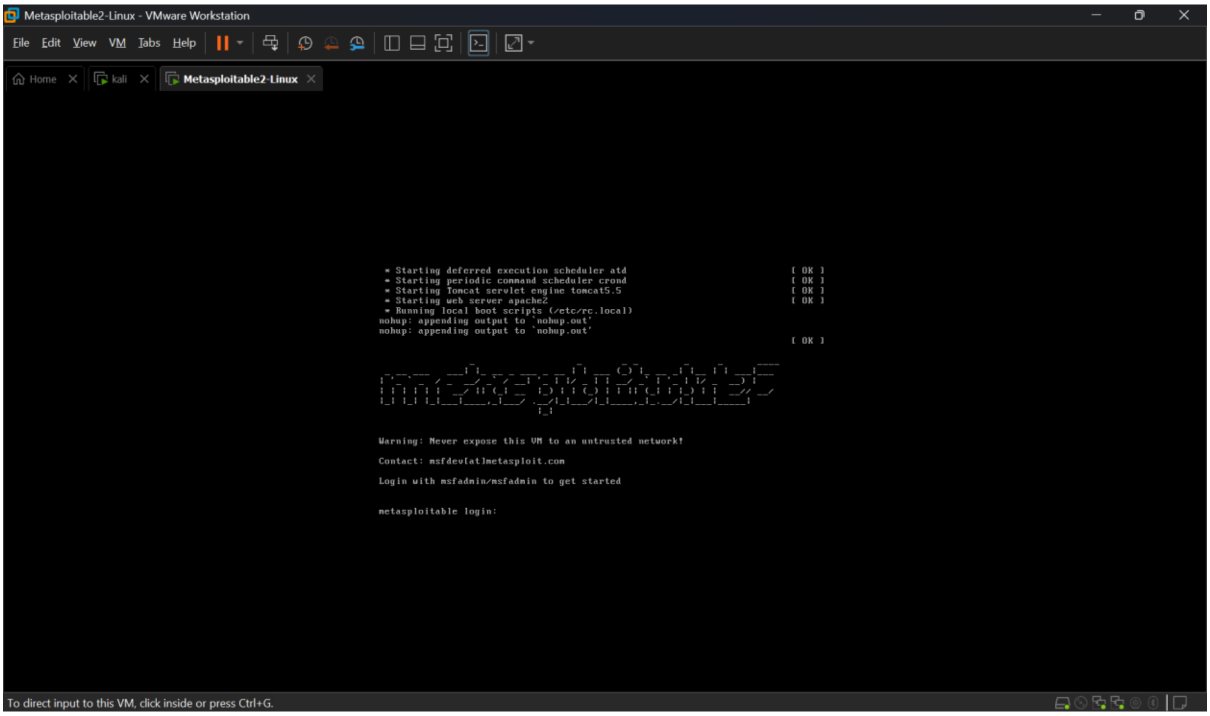
Screenshot

3.2 Metasploitable 2

Purpose

Metasploitable is an intentionally vulnerable Linux virtual machine designed for cybersecurity training and penetration testing practice.

Screenshot



4. Network Configuration

The virtual machines were connected using VMware virtual networking to allow communication between Kali Linux and Metasploitable.

IP Addresses

Machine	IP Address
Kali Linux	192.168.1.22
Metasploitable	192.168.180.131

Verification

The connectivity between the virtual machines was verified using the **ping** command.

Example:

```
ping 192.168.180.131
```

Expected Result:

- Successful replies received
- No packet loss

5. Wireshark Packet Capture

Wireshark was installed on Kali Linux to monitor network communication between the attacker and target machines.

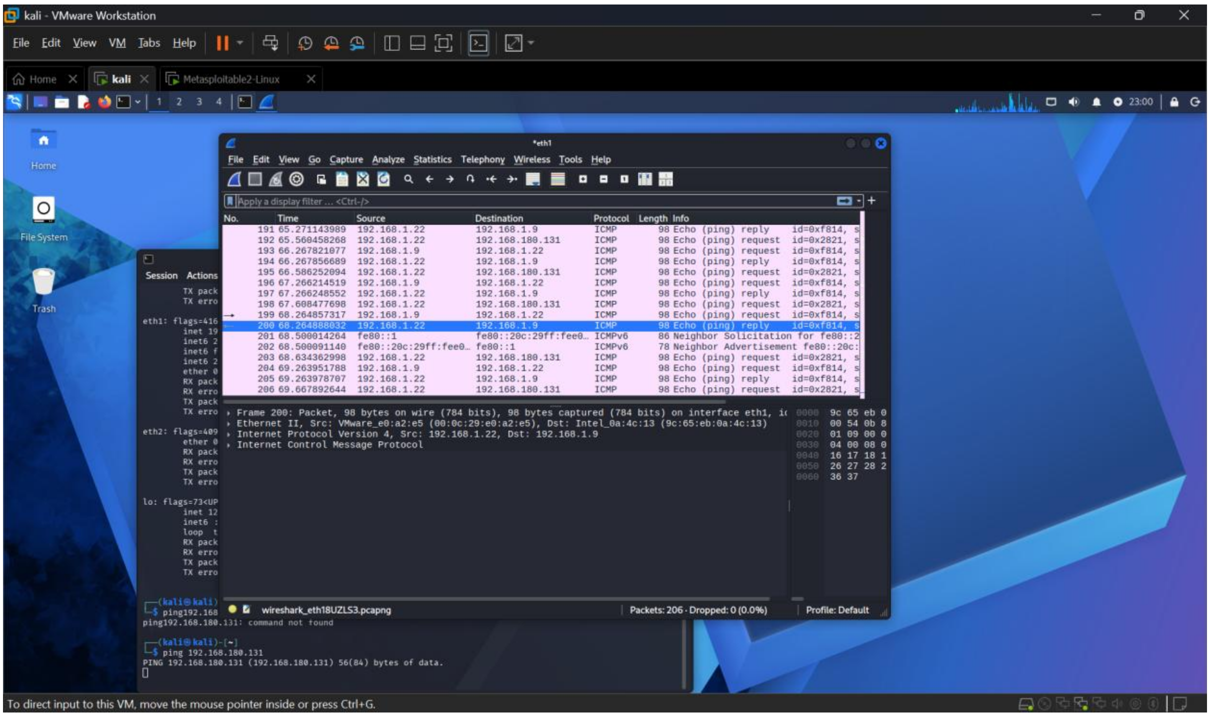
Test Performed

A ping request was sent from Kali Linux to Metasploitable while Wireshark captured the traffic.

Observed Packets

- ICMP Echo Request
- ICMP Echo Reply

Screenshot



6. Tools Verified

Tool	Status
Kali Linux	Working
Metasploitable	Working
Wireshark	Working
Ping Test	Successful
VMware Network	Working

7. Challenges Faced

- Mouse cursor visibility issue in Kali Linux after installation.
- Resolved by installing VMware Tools and enabling 3D acceleration.
- Updated Kali packages using:

  ```
  sudo apt update
  sudo apt full-upgrade
  ```

8. Learning Outcomes

After completing this lab setup, I learned:

- How to install and configure virtual machines.
- How to set up a cybersecurity testing environment.
- How to verify network connectivity between machines.
- How to capture packets using Wireshark.
- The role of vulnerable machines like Metasploitable in penetration testing.

9. Conclusion

The cybersecurity lab environment was successfully configured. Kali Linux and Metasploitable communicated over the virtual network, and Wireshark successfully captured network packets. This lab provides a safe environment for practicing ethical hacking, network analysis, and security testing without affecting real-world systems.